

Práctica en Git – Victor Alonso Camargo Flores

1. Configuración inicial

```
MINGW64:/c/Users/Usuario

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ git --version
git version 2.51.0.windows.1

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ git config --global user.name "AlonsoCamargo"

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ git config --global user.email "victorcamargoflores@gmail.com"

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ |
```

2. Crear repositorio

```
Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ mkdir git-practica

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ cd got-practica
bash: cd: got-practica: No such file or directory

Usuario@LAPTOP-L4M510NA MINGW64 ~ (main)
$ cd git-practica

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (main)
$ git init
Initialized empty Git repository in C:/Users/Usuario/git-practica/.git/
```

3. Primer archivo y primer commit

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ echo "# Proyecto de la practica con Git" > README.md

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git commit -m "Primer commit: agregar README"
[master (root-commit) f4e8130] Primer commit: agregar README
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

4. Revisar historial y el estado

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git status
On branch master
nothing to commit, working tree clean

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git log --oneline --graph
* f4e8130 (HEAD -> master) Primer commit: agregar README
```

5. Crear y modificar archivos

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ echo "print('Hola Git')" > app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git commit -m "Agregar archivo.py con saludo"
[master c75a083] Agregar archivo.py con saludo
1 file changed, 1 insertion(+)
create mode 100644 app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ echo "print('Modificado')" >> app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it
```

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git commit -m "Modificar app.py para imprimir mensaje nuevo"
[master 5c81d6a] Modificar app.py para imprimir mensaje nuevo
1 file changed, 1 insertion(+)
```

6. Deshacer cambios

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git checkout --app.py
error: unknown option 'app.py'
usage: git checkout [<options>] <branch>
       or: git checkout [<options>] [<branch>] -- <file>...

    -b <branch>          create and checkout a new branch
    -B <branch>          create/reset and checkout a branch
    -l                  create reflog for new branch
    --[no-]guess         second guess 'git checkout <no-such-branch>' (default)
    --[no-]overlay       use overlay mode (default)
    -q, --[no-]quiet     suppress progress reporting
    --[no-]recurse-submodules[=<checkout>]
                        control recursive updating of submodules
    --[no-]progress      force progress reporting
    -m, --[no-]merge     perform a 3-way merge with the new branch
    --[no-]conflict <style>
                        conflict style (merge, diff3, or zdiff3)
    -d, --[no-]detach    detach HEAD at named commit
    -t, --[no-]track[=(direct|inherit)]
                        set branch tracking configuration
    -f, --[no-]force     force checkout (throw away local modifications)
    --[no-]orphan <new-branch>
                        new unborn branch
    --[no-]overwrite-ignore
                        update ignored files (default)
    --[no-]ignore-other-worktrees
                        do not check if another worktree is using this branch
    -2, --ours           checkout our version for unmerged files
    -3, --theirs         checkout their version for unmerged files
    -p, --[no-]patch     select hunks interactively
    -U, --unified <n>   generate diffs with <n> lines context
    --inter-hunk-context <n>
                        show context between diff hunks up to the specified number of lines
    --[no-]ignore-skip-worktree-bits
                        do not limit pathspecs to sparse entries only
    --[no-]pathspec-from-file <file>
                        read pathspec from file
    --[no-]pathspec-file-nul
                        with --pathspec-from-file, pathspec elements are separated with NUL character

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git reset app.py
```

7. Crear y cambiar ramas

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git reset app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git branch nueva-funcionalidad

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git checkout nueva-funcionalidad
Switched to branch 'nueva-funcionalidad'

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (nueva-funcionalidad)
$ git checkout -b nueva-funcionalidad
fatal: a branch named 'nueva-funcionalidad' already exists

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (nueva-funcionalidad)
$ echo "print('Nueva funcionalidad')" >> app.py
```

8. Fusionar ramas (merge)

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (nueva-funcionalidad)
$ git checkout master
Switched to branch 'master'

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git merge nueva-funcionalidad
Updating 5c81d6a..884ca52
Fast-forward
 app.py | 1 +
 1 file changed, 1 insertion(+)
```

9. Conectar con un repositorio remoto

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git remote add origin https://github.com/AlonsoCamar28/Apps-Distribuidas_PracticaGuiada.git

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (master)
$ git branch -M main

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (main)
$ git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (12/12), 1.15 KiB | 196.00 KiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AlonsoCamar28/Apps-Distribuidas_PracticaGuiada.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Práctica Guiada

1. Preparar entorno

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (main)
$ mkdir git-conflictos

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica (main)
$ cd git-conflictos

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (main)
$ git init
Initialized empty Git repository in C:/Users/Usuario/git-practica/git-conflictos/.git/

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ echo "print('Hola desde main')" > app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git commit -m "Commit inicial con app.py"
[master (root-commit) b99dcb7] Commit inicial con app.py
1 file changed, 1 insertion(+)
create mode 100644 app.py
```

2. Crear y trabajar en feature-a

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git checkout -b feature-a
Switched to a new branch 'feature-a'

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (feature-a)
$ echo "print('Cambio desde feature A')" >> app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (feature-a)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (feature-a)
$ git commit -m "Feature A: Agregar mensaje en app.py"
[feature-a 0e6edfa] Feature A: Agregar mensaje en app.py
1 file changed, 1 insertion(+)

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (feature-a)
$ git checkout master
Switched to branch 'master'
```

3. Crear y trabajar en feature-b

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git checkout -b feature-b
fatal: a branch named 'feature-b' already exists

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ echo "print('Cambio desde feature B')" >> app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git add app.py
warning: in the working copy of 'app.py', LF will be replaced by CRLF the next time Git touches it

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git commit -m "Feature B: Agregar mensaje en app.py"
[master d86b1e6] Feature B: Agregar mensaje en app.py
1 file changed, 1 insertion(+)
```

4. Simular conflicto

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git merge feature-a
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master|MERGING)
$ git merge feature-b
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.
```

5. Resolver conflicto

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master|MERGING)
$ git add app.py

Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master|MERGING)
$ git commit -m "Resolver conflicto entre feature A y feature B"
[master 761be32] Resolver conflicto entre feature A y feature B
```

```
Usuario@LAPTOP-L4M510NA MINGW64 ~/git-practica/git-conflictos (master)
$ git log --oneline --graph --all
*   761be32 (HEAD -> master) Resolver conflicto entre feature A y feature B
| \
|  * 0e6edfa (feature-a) Feature A: Agregar mensaje en app.py
* | d86b1e6 Feature B: Agregar mensaje en app.py
| /
| * 5c8dd24 (feature-b) Feature B: agregar mensaje en app.py
| /
* ca0429a Feature A: Agregar mensaje en app.py
* b99dcb7 Commit inicial con app.py
```

